

M12 Protected Review & Repair

LOCKED: open each lesson section only after a genuine saved independent attempt. Review diagnoses and repairs; it is never the first solution path.

DE12-01 - Review after attempt

Expected proof: Your explanation keeps business row grain separate from physical partitions and says executors execute tasks while the driver coordinates the application.

Targeted repair: If everything feels like “a server,” start with one question: who plans, who executes, and what chunk of data does one task receive?

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-02 - Review after attempt

Expected proof: Expected: Spark version 4.2.0, local[2], shuffle partitions 4. Exact JVM/OS text may differ.

Targeted repair: If Java errors appear, verify Java ≥ 17 and JAVA_HOME before reinstalling Spark. If import fails, check that ``python -m pip`` and ``python`` are the same environment.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-03 - Review after attempt

Expected proof: Normal orders: 16 rows, quantity 31, amount 3770.00. order_id/customer_id/region/category are strings; qty integer; amount double.

Targeted repair: If numeric columns arrive as strings, check whether an explicit schema was applied and whether malformed data was silently turned null. Inspect bad records rather than casting blindly.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-04 - Review after attempt

Expected proof: Both routes must have three category rows and total amount 3770.00.

Targeted repair: If equality fails, compare filters, null behavior, aggregation aliases and ordering. Do not blame Spark SQL before comparing logical expressions.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-05 - Review after attempt

Expected proof: Your classification must explain why printing a DataFrame object is not the same as materializing all its rows.

Targeted repair: If you are unsure, ask whether the call returns a new logical DataFrame or asks Spark to compute/return/write a result.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-06 - Review after attempt

Expected proof: You can explain at least one cause of an Exchange and distinguish plan evidence from a tuning guess.

Targeted repair: If the plan is overwhelming, find the data source first, then work upward to filters, joins/exchanges and aggregates.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-07 - Review after attempt

Expected proof: Every choice includes its intended downstream effect and a correctness control that row count/amount stay 16/3770.00.

Targeted repair: If you only remember “coalesce is faster,” state whether you are increasing or decreasing partitions and whether balanced redistribution is required.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-08 - Review after attempt

Expected proof: Region totals must sum back to 3770.00 and grouped row count is four.

Targeted repair: If tuning discussion becomes guesswork, return to input size, partition sizes, task count, shuffle bytes and runtime evidence.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-09 - Review after attempt

Expected proof: Correct join output remains 16 rows, amount 3770.00, and zero unmatched customer segments.

Targeted repair: If a broadcast hint looks ignored, check join type, plan, table size/statistics and Spark support before forcing another hint.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-10 - Review after attempt

Expected proof: Good join: 16 rows / 3770.00. The bad dimension must produce more rows and a larger amount than the source.

Targeted repair: If a join has extra rows, do not add DISTINCT first. Test key uniqueness on each side and locate the key(s) with multiplicity.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-11 - Review after attempt

Expected proof: Your evidence includes key-frequency counts and an explanation of why equal partition count does not guarantee equal partition size.

Targeted repair: If you cannot see skew, sort key counts descending and compare the largest key with median/typical keys.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-12 - Review after attempt

Expected proof: Your explanation names at least two AQE behaviors and says correctness controls remain unchanged.

Targeted repair: If you describe AQE as “Spark automatically makes everything fast,” return to the exact documented adaptive rules and their configuration boundaries.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-13 - Review after attempt

Expected proof: You can state what is cached, why it is reused, how it is materialized, and when it is released.

Targeted repair: If caching seems to have no benefit, verify the same lineage is actually reused and compare runtime/storage metrics after materialization.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

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DE12-14 - Review after attempt

Expected proof: Full read remains 16 rows / 3770.00. East filter returns only East rows. Explain both business and physical partition meanings.

Targeted repair: If pruning is absent, verify the filter targets an actual partition column and inspect PartitionFilters rather than guessing from runtime alone.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-15 - Review after attempt

Expected proof: Built-in and UDF versions must return equivalent values before any performance comparison.

Targeted repair: If you reach for a UDF first, search current pyspark.sql.functions for an equivalent expression and compose built-ins where readable.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-16 - Review after attempt

Expected proof: Every local-inspection action states an expected maximum row count and why it is safe on the learner laptop.

Targeted repair: If you need "all rows to inspect," define the question you are debugging and reduce columns/rows first.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.

DE12-17 - Review after attempt

Expected proof: Tests are deterministic, order-aware where necessary, and fail when the intended contract is broken.

Targeted repair: If equality fails for distributed results, first decide whether row order is part of the contract. Sort by a stable key before comparing when appropriate.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

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DE12-18 - Review after attempt

Expected proof: Normal integrated controls: 16 order rows, 31 quantity, 3770.00 amount, 16-row customer join, zero unmatched customer segments, four region groups.

Targeted repair: If the integrated job fails, isolate in this order: environment/session -> schema/read -> quality -> join cardinality -> transformation -> action/write -> plan/tuning. Repair the smallest failed layer first.

Fresh micro-retry: change the key/filter/partition count/join side/input fixture or execution setting as appropriate, close Review, and reproduce the competency without copying the first answer.

Pass rule: code/result + validation + explain-back must agree. No numeric percentage infers mastery.